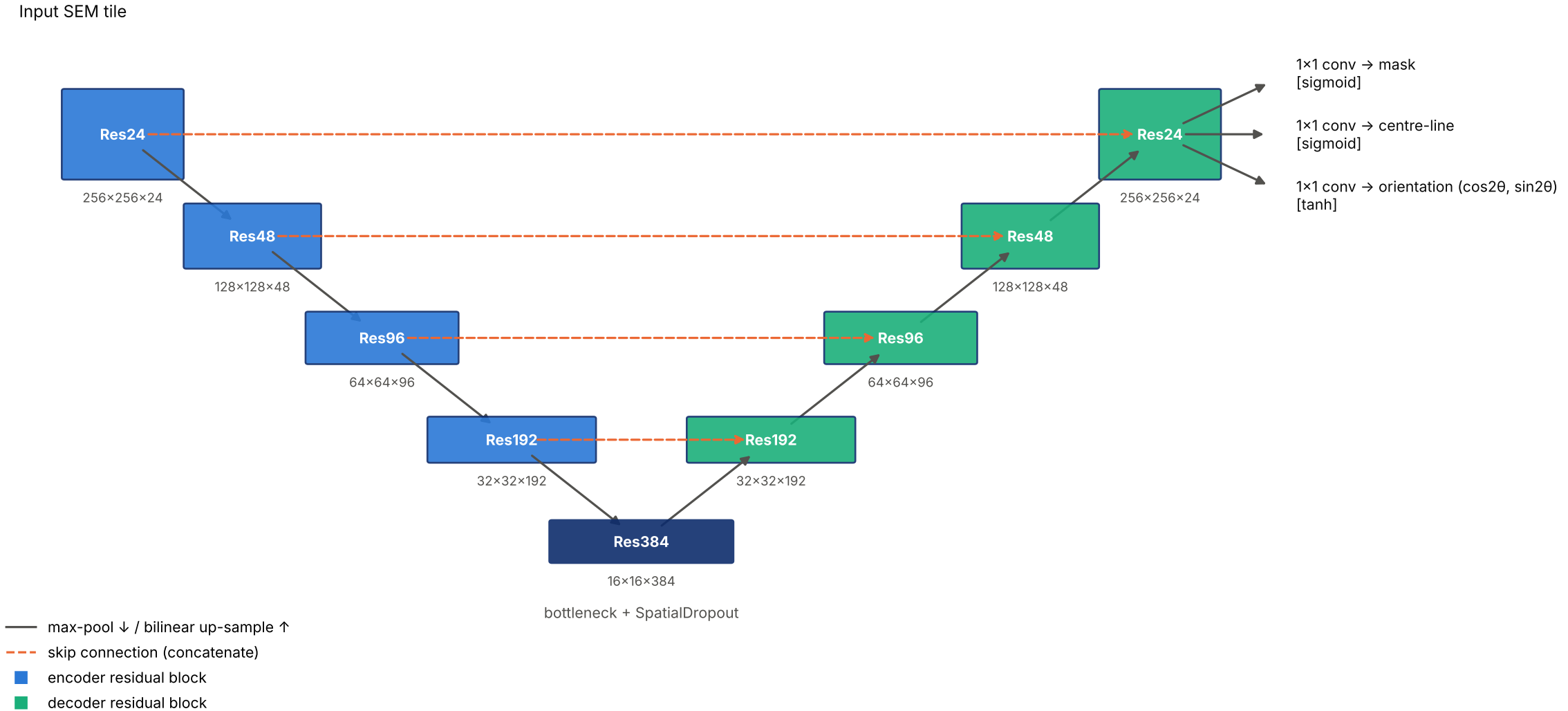
Input SEM tile
Res24
256×256×24
Res48
128×128×48
Res96
64×64×96
Res192
32×32×192
Res384
16×16×384
bottleneck + SpatialDropout
Res192
32×32×192
Res96
64×64×96
Res48
128×128×48
Res24
256×256×24
1×1 conv → mask [sigmoid]
1×1 conv → centre-line [sigmoid]
1×1 conv → orientation (cos2θ, sin2θ) [tanh]
max-pool ↓ / bilinear up-sample ↑
skip connection (concatenate)
encoder residual block
decoder residual block